DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-400-03 — Replace VALVE, SWING CHECK 3" P300 (4"-P300-P-1008/A2)

Area	A400 Pumps	**Task**	VR Valve replacement in kind
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	**P-102A** — P-102A centrifugal pump — E 164'-0" N -18'-6" EL 100'-9"		
Equipment	P-102A	**Lines**	4"-P300-P-1008
Isometrics	ISO-4"-P300-P-1008 (generated); owner's drawings ISO-P-1008		
Permits	General work permit		

Reason

Discharge check valve slamming; replace in kind.

Scope of work

1. Isolate per the isolation list, drain and vent 4"-P300-P-1008 between the isolations.
2. Unbolt and remove VALVE, SWING CHECK 3" P300 (4"-P300-P-1008/A2) at E 164'-0" N -16'-6" EL 104'-1/8".
3. Inspect both flange faces (SW-NDE-5); dress or report damage.
4. Install a new VALVE, SWING CHECK 3" P300 of the same size, class and face-to-face (CHECK3-P300), with new gaskets and studs, stem orientation as removed.
5. Tighten the studs to the stated preload in three passes, cross pattern (SW-WO-5).
6. Remove blinds, restore the line-up, leak test the two joints on start-up (SW-TEST-2).

Affected items: 4"-P300-P-1008/A2, 4"-P300-P-1008/A1, 4"-P300-P-1008/A4

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-NDE-5, SW-TEST-2, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1008	4"-P300-P-1008	4"-P300-P-1008	3	1 error(s), 4 warning(s) — see below

ISO-P-1008 (PDF) — the generated isometrics follow the ITP.

Drawing	Sheet	Level	Finding
ISO-P-1008	1	error	sheet 1: CONT'D ON ISO-P-1009 prints E 164'-3 5/16" N -17'-6" EL 107'-6" for E 164'-3 5/16" N -16'-6" EL 107'-6": the minus sign applies to the feet only; read as written the point is 12.00 in away and 12.00 in off where 4"-P300-P-1008 meets 4"-P300-P-1009 (E 164'-3 5/16" N -16'-6" EL 107'-6")
ISO-P-1008	2	warning	weld 4"-P300-P-1008/W14 (E 134'-8 11/16" N 2'-0" EL 116'-6", sheet 2) has no label
ISO-P-1008	3	warning	W15 (4"-P300-P-1008/W15, sheet 3) is labelled again on sheet 2; it is 4.0 in from the nearest break
ISO-P-1008	2	warning	branch weld 3/4"-P300-SP-7024/W1 (E 136'-0" N 2'-0" EL 116'-9 5/16", sheet 2) has no label
ISO-P-1008	2	warning	branch weld 3/4"-P300-I-7008/W1 (E 135'-0" N 2'-0" EL 116'-9 5/16", sheet 2) has no label

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
P-102A	equipment	P-102A centrifugal pump		E 164'-0" N -18'-6" EL 100'-9"
4"-P300-P-1008/A2	item	VALVE, SWING CHECK 3" P300	4"-P300-P-1008	E 164'-0" N -16'-6" EL 104'-1/8"
4"-P300-P-1008/A1	item	GASKET 3" P300 COMPRESSED SHEET	4"-P300-P-1008	E 164'-0" N -16'-6" EL 103'-4-15/16"
4"-P300-P-1008/A4	item	GASKET 3" P300 COMPRESSED SHEET	4"-P300-P-1008	E 164'-0" N -16'-6" EL 104'-7-5/16"
P-102A-D	nozzle	Nozzle P-102A-D (discharge, 3" P300) on P-102A		E 164'-0" N -16'-6" EL 103'-4-7/8"
4"-P300-P-1008/A5	item	VALVE, GATE, WEDGE, OS&Y 3" P300	4"-P300-P-1008	E 164'-0" N -16'-6" EL 105'-2-3/16"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	equipment	P-102A-D	4"-P300-P-1008	Line end at nozzle P-102A-D: the equipment is out of service, drained and isolated under the shutdown isolation			
2	valve	4"-P300-P-1008/A5	4"-P300-P-1008	Close and lock VALVE, GATE, WEDGE, OS&Y 3" P300 (4"-P300-P-1008/A5)	E 164'-0" N -16'-6" EL 105'-2-3/16"		
3	blind	4"-P300-P-1008/A2	4"-P300-P-1008	Fit a line blind between 4"-P300-P-1008/A5 and VALVE, SWING CHECK 3" P300 (4"-P300-P-1008/A2), work side	E 164'-0" N -16'-6" EL 103'-5"		

Parts

Description	Qty	Replaces
VALVE, SWING CHECK 3" P300	1	4"-P300-P-1008/A2
GASKET 3" P300 COMPRESSED SHEET	1	4"-P300-P-1008/A1
STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	1	4"-P300-P-1008/A3
GASKET 3" P300 COMPRESSED SHEET	1	4"-P300-P-1008/A4
STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	1	4"-P300-P-1008/A6

Inspections

Id	Inspection	Criteria	Basis
WO-400-03-I1	Flange faces of both joints before the new gaskets go in	No radial scratch across the sealing face (SW-NDE-5)	SW-NDE-5

Bolt-up (SW-WO-5)

Joint	Studs	Preload per stud (lbf)	Passes	Done ✓
4"-P300-P-1008/A1	4 × 0.875" studs (4"-P300-P-1008/A3)	27,059	3	
4"-P300-P-1008/A4	4 × 0.875" studs (4"-P300-P-1008/A6)	27,059	3	

Test — Service leak test

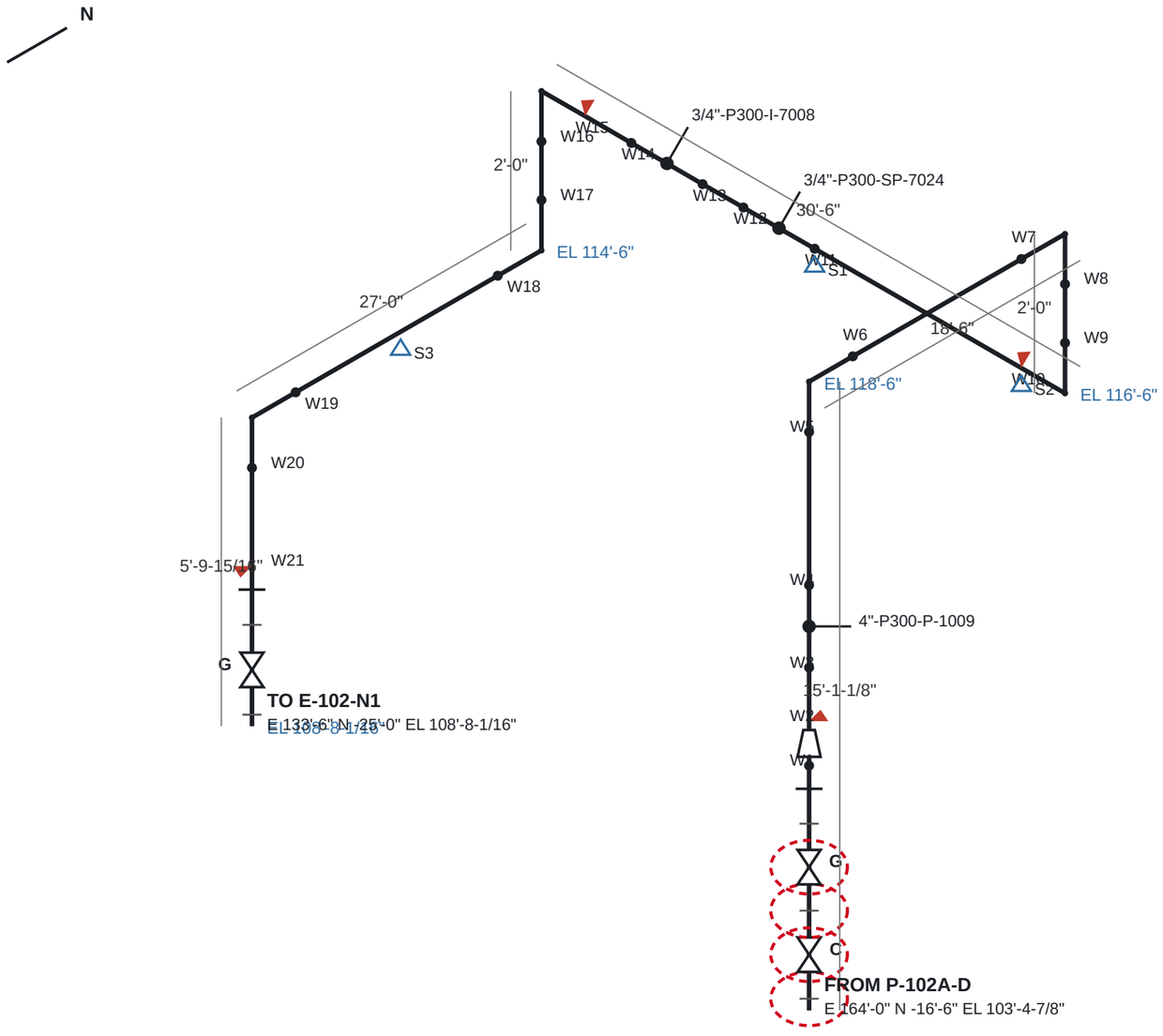
Description	Both remade joints checked at operating pressure on start-up
Basis	SW-TEST-2

Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Valve 4"-P300-P-1008/A2 removed; flange faces inspected	SW-NDE-5	No radial scratch across the sealing face	Flange face inspection record	W	W	—				
3	New valve verified: type, size, class, face-to-face and tag	CHECK3-P300	Matches the parts list and the line class	Material receiving record	R	R	—				
4	New gaskets fitted, studs tightened to preload in three passes	SW-WO-5	Preload per stud as stated; three passes, cross pattern	Bolt-up record per joint	W	W	—				
5	Service leak test of the remade joints on start-up	SW-TEST-2	No leak at either joint	Leak test record	W	W	—				
6	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
7	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

ISOMETRIC — 4"-P300-P-1008 — Replace VALVE, SWING CHECK 3" P300 (4"-P300-P-1008/A2)



BILL OF MATERIALS

1	GASKET 3" P300 COMPRESSED SHEET	3
2	VALVE, SWING CHECK 3" P300	1
3	STUD SET 4 × 0.875" × 5-3/4" WITH 2 NUTS, FOR 3" P300	3
4	VALVE, GATE, WEDGE, OS&Y 3" P300	1
5	FLANGE WELD-NECK 3" P300 RAISED FACE	1
6	REDUCER CONCENTRIC 4" × 3" P300	1
7	PIPE 4" P300 BEVELLED ENDS	87'-8-7/16"
8	TEE EQUAL 4" P300	1
9	ELBOW 90° R = 5/3 OD 4" P300	6
10	TEE REDUCING 4" × 3/4" P300	2
11	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300	2
12	FLANGE WELD-NECK 4" P300 RAISED FACE	1
13	GASKET 4" P300 COMPRESSED SHEET	2
14	VALVE, GATE, WEDGE, OS&Y 4" P300	1

WELDS 21 (4 FIELD) · SUPPORTS 3 · SPOOLS 5

W1 W2F W3 W4 W5 W6 W7 W8 W9 W10F
W11 W12 W13 W14 W15F W16 W17 W18 W19 W20
W21F

4"-P300-P-1008

CLASS H300 — Hydrocarbon process
SERVICE P · 4" · P300 · DESIGN 300 psi
P-102A-D → E-102-N1
PIPE 87'-8-7/16" · WEIGHT 1192 lb

DWG ISO-4"-P300-P-1008 REV 0 WO-400-03

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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